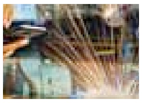


the field. Some proximity sensors also utilise ultrasonic waves to calculate the distance between the object and themselves. Proximity sensors see some unique use cases in IoT devices. They are sometimes used in malls to detect the distance between the customer and the object they may be interested in buying. When the customer comes closer, the sensor triggers a notification (audio, visual, or digital) which lets the customer know about any discounts or special offers relating to the product. They are also used in parking lots to indicate parking availability.

PRESSURE SENSORS

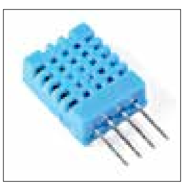
A pressure sensor is a device that measures the pressure (the force needed to stop a fluid from expanding) in liquids or gases. These are one of the most popular IoT sensors primarily due to their industrial applications, since industries are increasingly adopting IoT connectivity. Now, there are numerous devices that rely on liquid or other forms of pressure. The pressure sensor monitors all these systems and devices that are pressure-propelled. If there is any deviation, however slight, from the standard pressure range, the device then notifies the connected system about the deviations. These are typically used to check for water leaks in residential or commercial areas. They are also useful in manufacturing processes, maintenance of whole water systems, heating systems and more.

OPTICAL SENSORS

Optical sensors measure the physical quantity of light rays coming from any source (the sun, bulbs, fire, etc) and then convert this into electrical signals which can be read by either the user or an electronic instrument. Optical sensors are generously used in an array of different IoT devices or applications. They are used in healthcare, environment monitoring, energy, aerospace, and other industries. The biggest example of optical sensors being used in cutting edge IoT technology is automated cars. The induction of optical sensors in this space has allowed smart parking applications, object recognition, and light detection in autonomous cars. These sensors are also being utilised in smart city projects for security systems, face recognition, city infrastructure monitoring, and more. They are also being used in smartphones to power in-display fingerprint security systems.

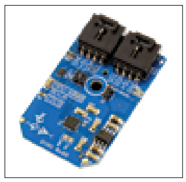
WATER QUALITY SENSORS

While most of us in urban areas take water quality for granted, many people in rural areas do not have access to purified and clean water. This is where water sensors come into play. Water quality sensors are tasked with detecting a range of parameters that determine the quality of water. These devices are implemented across the world in water filtration and distribution systems. They are used widely in smart systems that monitor the amount of contaminants in water and issue notices/warnings if the water quality is too poor. Water quality sensors also help monitor sea pollution levels by incorporating other sensors such as pH and OPR to measure the acid-base balance or the number of suspended solids in the sea water. These sensors are also being utilised in modern swimming pools that monitor the germs and bacteria in the water and start cleaning activities automatically when needed.



HUMIDITY SENSORS

Humidity sensors decipher the amount of water vapour and other gases in the atmosphere. This is also referred to as the 'relative humidity' or RH levels. These sensors are also commonly implemented where temperature sensors are usually found. This is because manufacturing processes often require pristine working conditions and keeping the humidity in check can often ensure smooth running of manufacturing processes. The sensors can detect change in humidity almost instantly which allows quick intervention if the manufacturing process is exceptionally susceptible to humidity. Humidity sensors can also be used to aid or trigger heating, ventilating, and air condition systems in industries and residential areas. In smart homes, humidity sensors can help keep track of the dryness of the air. Devices fitted with humidity sensors collect real-time humidity data and transfer the data to the network. After this, actionable commands are generated which can adjust the HVAC systems or activate a humidifier in smart homes.



ACCELEROMETERS AND GYROSCOPES

Most of you will be acquainted with gyroscopes and accelerometers since they are commonly found in smartphones. These are the sensors that are able to sense if your phone is in portrait mode or landscape mode and are also used in mobile games such as PUBG Mobile to aid movement (if the option is turned on). These sensors are just as popular in IoT applications too. Most of the time, accelerometers and gyroscopes are shipped together as one unit, however, there is a difference between the two – gyroscopes make use of the Earth's gravity to determine orientation while accelerometers are designed to measure non-gravitational acceleration. In IoT, accelerometers are used in smart pedometers and monitoring driving fleets. They can also be used as an anti-theft system which alerts the system if an object that should be stationary is moved. Gyros are commonly used to determine the moving direction of a GPS-tracked object. It can also be used to detect wind direction and are used in clean energy applications.